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SUPERSEDING
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August 2, 1965

FEDERAL SPECIFICATION

ALUMINUM ALLOY SHEET, PAINTED (FOR EXTERIOR USE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for aluminum alloy sheet coated with a baked on organic finish.

1.2 Classification.

1.2.1 *Classes.* Aluminum alloy painted sheet shall be of the following classes, as specified (see 6.2):

Class a—High formability.
Class b—Moderate formability.

1.2.2 *Finishes.* Aluminum alloy painted sheet shall be of the following finishes, as specified (see 3.4.7 and 6.2):

Finish 1—Gloss.
Finish 2—High semigloss.
Finish 3—Standard semigloss.
Finish 4—Low semigloss.
Finish 5—Eggshell.
Finish 6—Flat.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specification:

UU-T-106—Tape, Pressure-Sensitive Adhesive, Masking, Paper.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-C-5541—Chemical Films for Aluminum and Aluminum Alloys.

Military Standards:

MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-649—Aluminum and Magnesium Products, Preparation for Storage and Shipment of.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society For Testing And Materials (ASTM) Publications:

D 523—Test for Specular Gloss.
D 659—Method of Evaluating Degree of Resistance to Chalking of Exterior Paints.
D 714—Method of Evaluating Degree of Blistering of Paints.
D 1400—Measurement of Dry Film Thickness of Nonmetallic Coatings of Paint, Varnish, and Lacquer, and Related Products Applied on a Nonmagnetic Metal Base.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa., 19103.)

3. REQUIREMENTS

3.1 Material. The aluminum alloy sheet used as the base material shall be as specified (see 6.2).

3.2 Dimensions. The thickness, length, and width of the sheets to be covered with the coating material shall be as specified (see 6.2), and shall conform to the tolerances of the applicable specifications.

3.3 Pretreatment. Prior to application of the baked organic finish, aluminum alloy sheet shall receive a chemical conversion coating conforming to MIL-C-5541.

3.4 Baked organic finish. Unless otherwise specified (see 6.2), the aluminum sheet or fabricated product shall have a baked organic finish on one side only. The edges of the sheet need not be coated. When the baked organic finish is applied on one side only, a wash coat of enamel shall be applied to the reverse side.

3.4.1 *Thickness.* The minimum dry film thickness when tested in accordance with 4.4.1 shall be 0.8 mil.

3.4.2 *Hardness.* The dry film shall not rupture when tested in accordance with 4.4.2.

3.4.3 *Impact resistance.* Finish 1 coated aluminum alloy sheet and fabricated products in tempers other than —0 (annealed) shall withstand direct and reverse impact when tested in accordance with 4.4.3.

3.4.4 *Film adhesion.* No removal of the film shall occur when tested in accordance with 4.4.4.

3.4.5 *Film flexibility.* Classes a and b, respectively, shall be capable of passing the flexibility test of 4.4.10 with no evidence of cracking, flaking, or spalling.

3.4.6 *Color uniformity.* The color uniformity shall be commercially constant when tested in accordance with 4.4.5.

3.4.7 *Specular gloss.* The specular gloss shall conform to required gloss ranges of table I or to special gloss ranges as specified (see 6.2), when measured at a 60° angle in accordance with 4.4.6.

TABLE I. *Specular gloss at 60° angle*

Finish	Specular gloss	
	Minimum	Maximum
1	80	—
2	60	79
3	40	59
4	20	39
5	8	19
6	—	7

3.4.8 *Humidity resistance.* The painted metal shall be capable of withstanding the test specified in 4.4.7 with no formation of blisters to an extent greater than illustrated for "Few Blisters of No. 8 Size" in figure 4 of ASTM D 714. There also shall be no loss of adhesion, specular gloss, color, or color uniformity.

3.4.9 *Salt spray resistance.* The painted metal shall be capable of withstanding the salt spray test of 4.4.8 with a maximum undercutting of the film from the scored line of 1/16 inch.

3.4.10 *Chemical resistance.* The painted

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metal shall be capable of withstanding the test specified in 4.4.9 with no loss of adhesion or gloss and no color change or staining.

3.4.11 *Accelerated weathering.* The painted metal shall be capable of withstanding the requirements of 3.4.11.1 and 3.4.11.2.

3.4.11.1 *White.* When exposed as specified in 4.4.11, the painted metal shall show no film defects other than very slight yellowing (yellowness of delta N of 0.1 maximum) when tested in accordance with method 6131 of Fed. Std. No. 141 and no more than slight chalking, (degree of chalking No. 6 maximum, fig. 1 of ASTM D 659).

3.4.11.2 *Colors.* When exposed as specified in 4.4.11, the coating shall show no more than very slight color change or chalking, and no other film defects.

3.4.12 *Color standards.* Unless other provisions are made by the bureau, department, or other Government activity concerned, the contracting officer shall arrange to furnish the supplier with the necessary reference color standards (see 6.2).

3.5 **Workmanship.** The baked organic coating shall be commercially smooth and substantially free from flow lines, streaks, blisters, cracks, scratches, pinholes, or flaked areas.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling.**

4.2.1 *Lot.* A lot shall consist of an identifiable quantity of painted sheet of the same base alloy, temper, nominal thickness, and color subjected to inspection at one time.

4.2.2 *Sampling for coating tests and examination.* Twelve samples shall be taken at random from each 1,000 to 10,000 pounds of painted sheet, which shall comprise a lot for the testing or examination specified in table II. For lots of less than 1,000 pounds, two samples shall be taken at random for testing or examination.

TABLE II. *Testing or examination*

Test	Requirement paragraph	Test paragraph
Thickness	3.4.1	4.4.1
Hardness	3.4.2	4.4.2
Impact resistance	3.4.3	4.4.3
Film adhesion	3.4.4	4.4.4
Color uniformity	3.4.6	4.4.5
Specular gloss	3.4.7	4.4.6
Flexibility	3.4.5	4.4.10

4.3 **Examination.**

4.3.1 *End item.* Samples selected in accordance with 4.2.2 shall be examined for defects listed in table III. The acceptable quality levels (AQL's) for defects shall be 2.5 percent defective for major defects, and 4.0 percent defective for minor defects.

TABLE III. *Classification for defects, end item*

Defects	Major	Minor
Not smooth	X	
Not free from flow-lines	X	
Not free from streaks or blisters	X	
Not free from scratches	X	
Not free from cracks or chips	X	
Not free from pinholes or flaked areas	X	
Damage or defects not affecting function or serviceability		X
Damage or defects affecting function or serviceability	X	

4.3.2 *Inspection of preparation for delivery.* An inspection shall be made to determine that the packaging, packing, and marking requirements comply with section 5.

4.4 **Tests.**

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4.4.1 *Thickness.* The dry film thickness shall be determined in accordance with method 6183 of Fed. Std. No. 141 or method A in ASTM D 1400 (see 3.4.1).

4.4.2 *Hardness.* The dry film hardness shall be determined by the use of an HB Eagle Turquoise pencil. The wood shall be stripped from the pencil leaving the full diameter of the lead. The end of the lead shall be flattened by using fine grit sand or emory paper so that the end is 90° to the pencil axis. The pencil shall be held at a 45° angle to the film surface and pushed forward approximately 1/4 inch using as much downward pressure as can be applied without breaking the pencil lead (see 3.4.2).

4.4.3 *Impact resistance.* The sample shall be tested at room temperature with a Gardner variable impact tester or equivalent, with a 5/8 inch mandrel. The initial impact force shall be 40 inch-pounds; however, if the metal is ruptured, then the force shall be decreased by 5 inch-pound increments until no metal fracturing occurs. Minute fracturing of the film is permissible, but when masking tape conforming to UU-T-106, 3/4 inch wide, is applied to the deformed coated area as per 4.4.4 and pulled off sharply, no film shall be removed (see 3.4.3).

4.4.4 *Film adhesion.* The dry film adhesion shall be tested making 10 parallel cuts through the film and down to the metal approximately 1/16 inch apart, by means of a knife or similar instrument. Ten similar cuts shall be made at 90° to, and crossing the first 10 cuts. A 3/4 inch wide strip of masking tape conforming to UU-T-106 shall be applied to the area, adhesive side down. The tape shall be pressed down using two passes of a 4-1/2 pound rubber-covered roller approximately 3-1/2 inches in diameter by 1-3/4 inches in width, the surface of which has a durometer hardness value within the range of 70 to 80. The roller shall be moved at the rate of approximately 1 inch per second using the weight of the roller only. The tape shall be removed in one abrupt motion with the pull exerted at approximately a 90° angle to the panel. No film shall be removed (see 3.4.4).

4.4.5 *Color uniformity.* The color uniformity shall be checked visually under a uniform light source such as a McBeth Daylight lamp (see 3.4.6).

4.4.6 *Specular gloss.* The specular gloss shall be determined in accordance with ASTM D 523 (see 3.4.7).

4.4.7 *Humidity resistance.* Humidity resistance shall be tested in accordance with method 6201 in Fed. Std. No. 141, exposing the samples to 100 percent relative humidity for 1,000 hours (see 3.4.8).

4.4.8 *Salt spray resistance.* Salt spray resistance shall be tested in accordance with method 6061 in Fed. Std. No. 141, exposing the specimen for 1,000 hours to a fog spray of 5 percent sodium chloride solution (see 3.4.9).

4.4.9 *Chemical resistance.* Resistance to chemicals shall be tested on separate samples immersed in mineral spirits, 0.5 percent tide solution, 2.0 percent trisodium phosphate and 2.0 percent ivory soap solution, then removed, rinsed, and allowed to recover for 24 hours (see 3.10).

4.4.10 *Flexibility.* A 2 by 12 inch specimen shall be bent cold, once through 180°, at room temperature. Determination of cracking, flaking, or spalling shall be by visual examination without magnification (see 3.4.5).

4.4.10.1 *Class a.* The specimen shall be bent over two thicknesses of the material.

4.4.10.2 *Class b.* The specimen shall be bent over a 1/2 inch diameter mandrel.

4.4.11 *Accelerated weathering.* Accelerated weathering shall be tested either in accordance with method 6151 of Fed. Std. No. 141 for 500 hours in an Atlas Type XW Sunshine Arc Weatherometer or equivalent, or in accordance with method 6152 of Fed.

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Std. No. 141 for 1,000 hours in an Atlas Type DMC enclosed Violet Carbon Arc Weatherometer or equivalent (see 3.4.11). Each 2 hour cycle of operation shall consist of 102 minutes of light and 18 minutes of light and demineralized water.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. All material shall be preserved and packed in accordance with the requirements of MIL-STD-649. Unless otherwise specified (see 6.2), material shall be packed in accordance with level C.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The aluminum alloy painted sheets covered by this specification are intended for fabrication into such building products as siding, gutters, and downspouts. For lap siding, gutters, and downspouts requiring sharp bends, class a sheets should normally be specified, while class b sheets should be satisfactory for corrugated siding, flat panels, or items with sharp bends painted after fabrication.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Class and finish (see 1.2, 6.1, and 6.3).
(*c*) Alloy and temper of base metal (see 3.1 and 6.3).
(*d*) Dimensions (see 3.2).
(*e*) Number of sides to be coated (see 3.4).
(*f*) Special gloss range, if required (see 3.4.7).
(*g*) Color of coating (see 3.4.12).
(*h*) Selection of applicable level of packing required, if other than level C (see 5.1).

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